

CLINICAL PHARMACOKINETICS: Applications



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BPM1 PHARMACOLOGY
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Objectives

- SOM.MK.1.BPM1.1.FTM.4.PHAR.0034
Use pharmacokinetic concepts and principles to design a rational dosage regimen. Define maintenance dose. Describe the concept of steady-state with regard to plasma drug concentrations.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0035
Given a drug's clearance, calculate the maintenance dose needed to maintain an average steady-state drug level using either a continuous infusion or a fixed-dose, fixed-time interval regimen. Given a drug's clearance, and the dosing rate, calculate the steady-state drug plasma level achieved.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0036
Describe different dosage regimens.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0037
Define loading dose. Calculate a loading dose to achieve a target plasma drug level.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0038
Given the half-life of a drug, calculate the accumulation factor. Calculate peak and trough concentrations at steady state for a drug administered by multiple IV injections.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0039
Plot the time course of plasma drug concentration following different routes of administration. Compare and contrast the time course of plasma concentration for an immediate-release oral formulation and a slow-release oral formulation

DESIGNING A RATIONAL DOSAGE REGIMEN

- A dosage regimen is a plan for drug administration over a period of time.
- It should result in therapeutic levels of the drug in blood without exceeding the minimum toxic concentration.

DESIGNING A RATIONAL DOSAGE REGIMEN

- To maintain the plasma concentration within a specified range over long periods of therapy, a schedule of **maintenance doses** is used.
- If it is necessary to achieve a target plasma level rapidly, a **loading dose** is used to load the V_d with the drug.

DESIGNING A RATIONAL DOSAGE REGIMEN

- The dosing plan is based on knowledge of:
 - Therapeutic window
 - Clearance
 - V_d

THERAPEUTIC WINDOW

- Determines the range of plasma levels acceptable when designing a dosing regimen.

MAINTENANCE DOSE

- In most clinical situations drugs are administered to maintain a **steady state** of drug in the body:
- Just enough drug is given in each dose to replace the drug eliminated since the preceding dose.

MAINTENANCE DOSE

- At steady state (SS) the dosing rate (“rate in”) must equal the rate of elimination (“rate out”).

$$\text{Dosing rate}_{ss} = \text{rate of elimination}_{ss} = \frac{CL \times C}{F}$$

MAINTENANCE DOSE

- Thus, if the desired target concentration is known, the clearance in that patient will determine the dosing rate.
- If intermittent doses are given, the maintenance dose is calculated with:

Maintenance dose = dosing rate x dosing interval

DOSAGE REGIMENS

CASE 1. CONTINUOUS IV INFUSION

- With continuous IV infusion, the rate of drug entry in the body is constant.
- The plasma concentration of drug rises until the rate of drug eliminated from the body balances input rate.
- A steady-state is achieved in which drug concentration remains constant.

CASE 1. CONTINUOUS IV INFUSION

- Rate of elimination = $CL \times C$
- C_{ss} is achieved when rate of drug elimination is equal to rate of administration.

$$C_{ss} = \frac{\text{Infusion rate}}{CL}$$

CASE 1. CONTINUOUS IV INFUSION

- CL is constant for drugs showing first-order kinetics.
- Therefore the steady-state plasma concentration (C_{ss}), is directly proportional to the infusion rate.
- If infusion rate is doubled, steady-state plasma concentration is doubled.

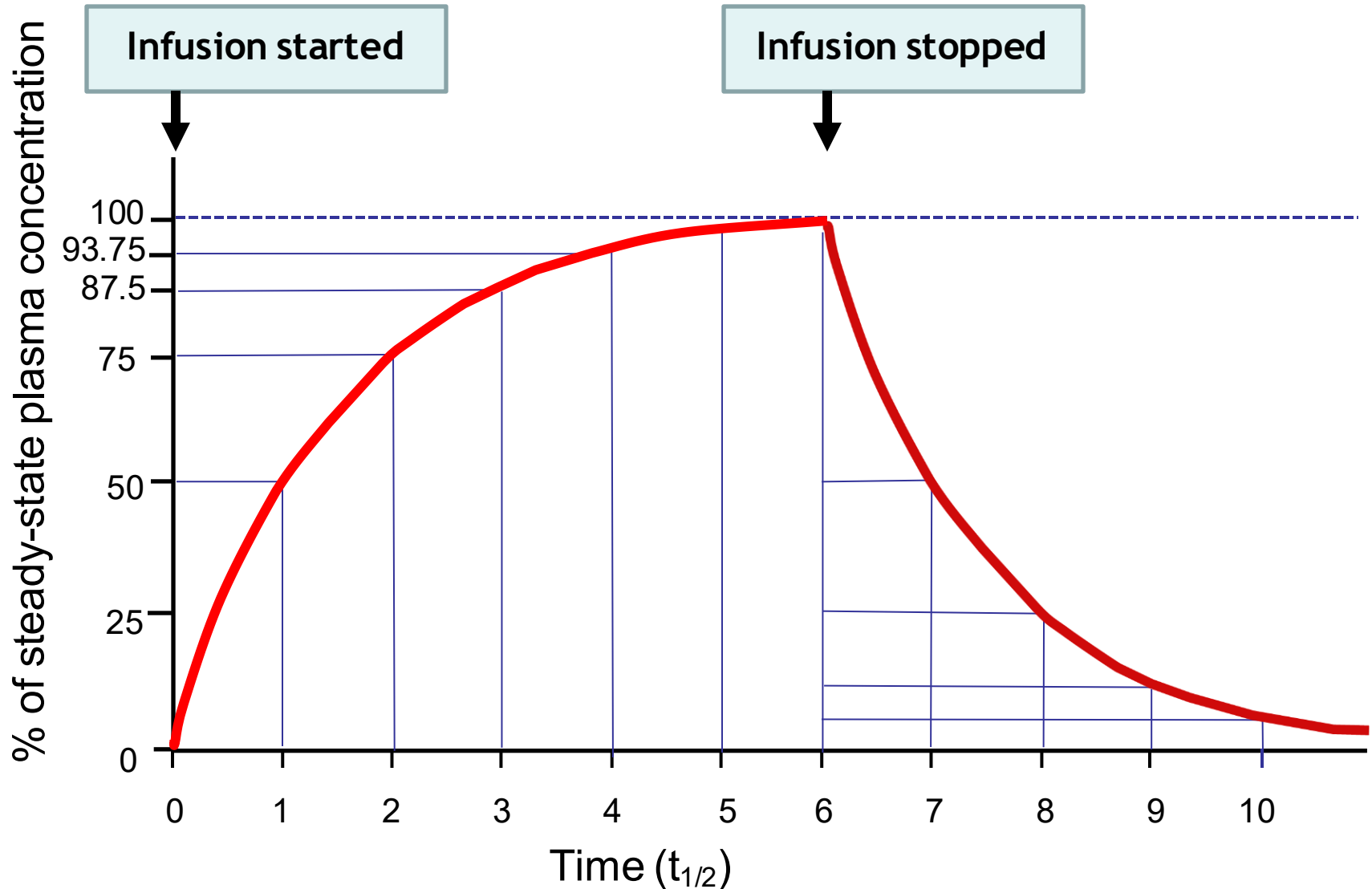
CASE 1. CONTINUOUS IV INFUSION

- The rate of approach to steady state follows first-order kinetics.
- 50% of the final steady-state (C_{ss}) is attained after 1 half-life.
- 75% after 2 half-lives.
- 87.5% after 3 half-lives
- 93.75% after 4 half-lives
- It can be assumed that steady state is reached after about 4 half-lives.

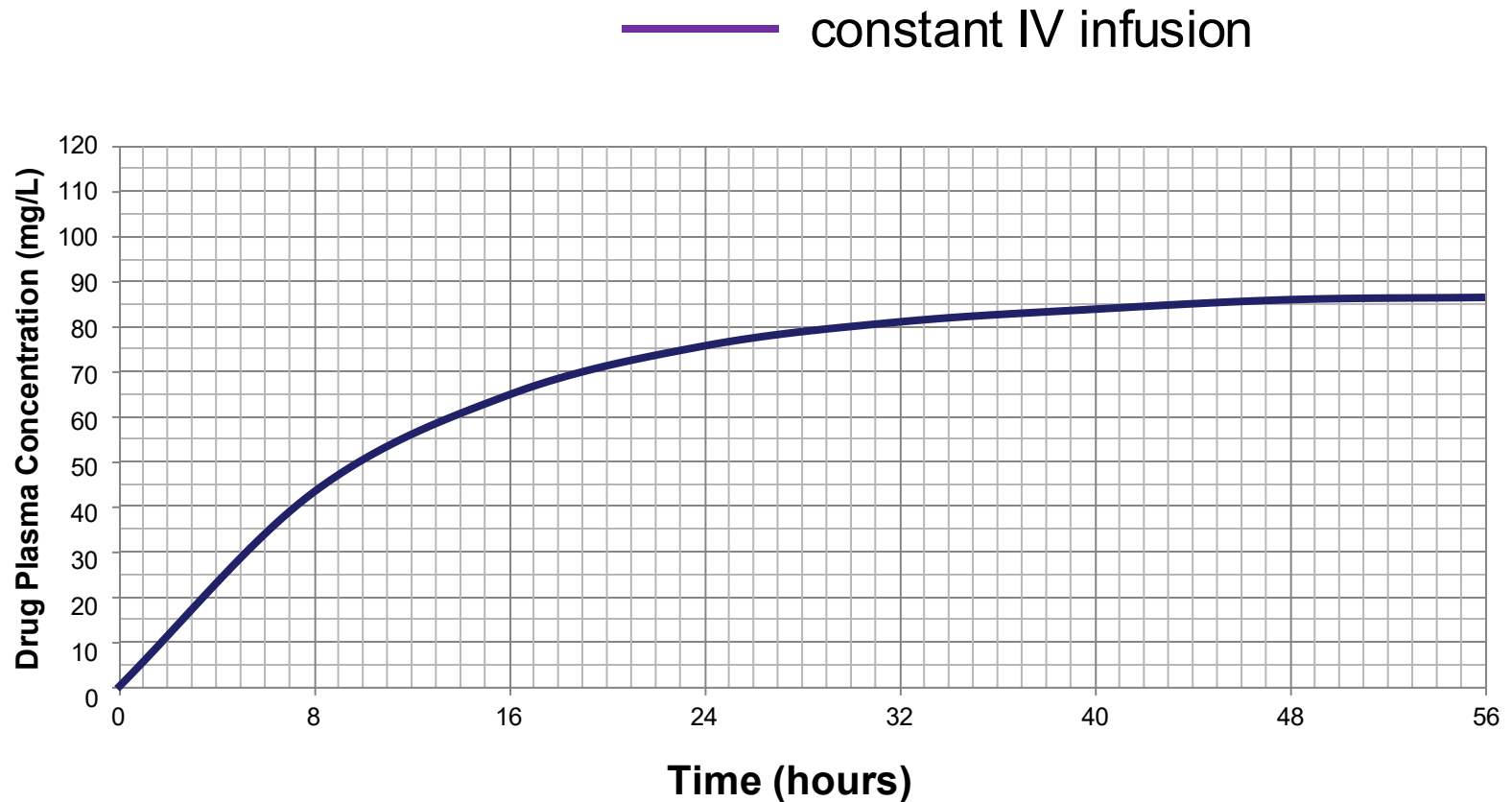
CASE 1. CONTINUOUS IV INFUSION

- When infusion is stopped, plasma concentration declines to zero with same time course observed in approaching steady state.

CASE 1. CONTINUOUS IV INFUSION



CASE 1. CONTINUOUS IV INFUSION



LOADING DOSE

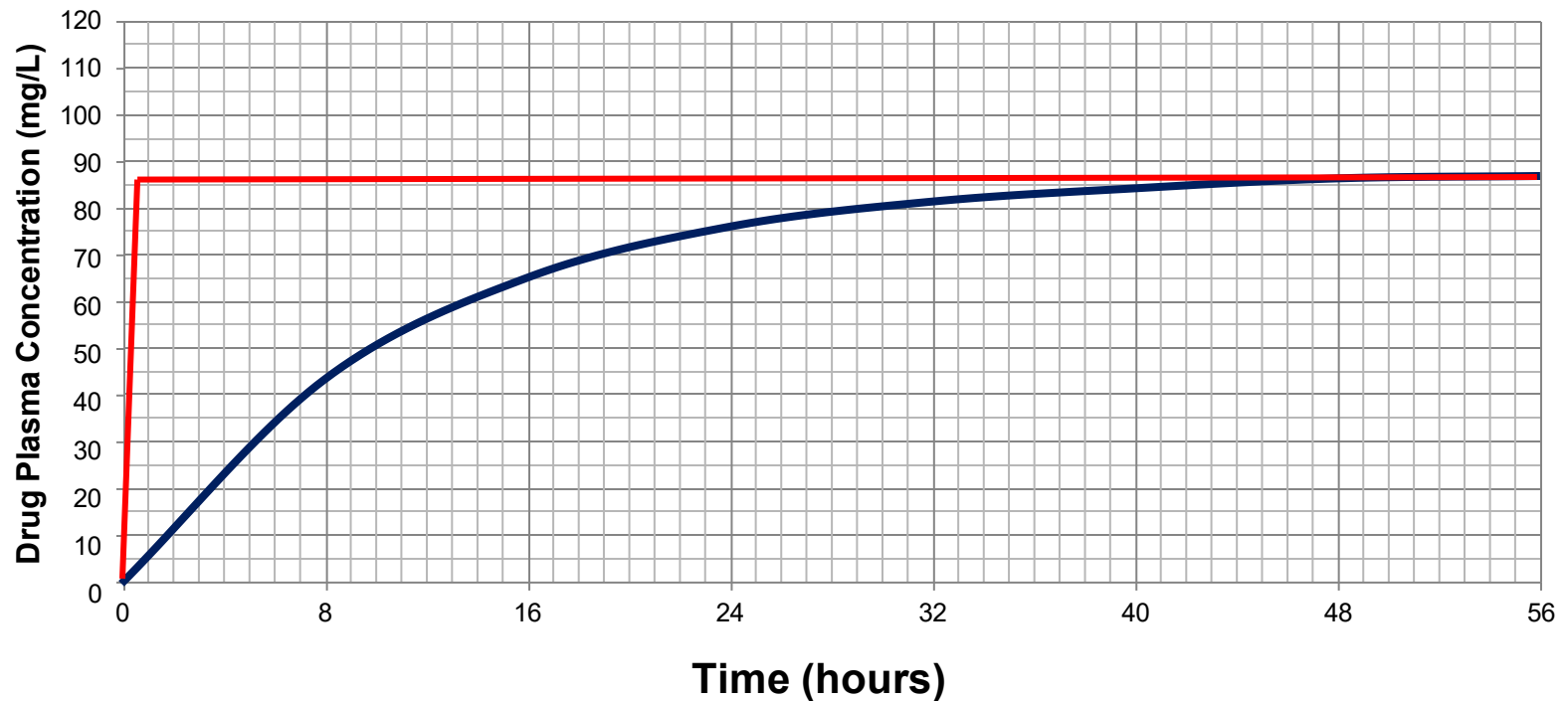
- If the therapeutic concentration must be achieved rapidly and the V_d is large, a large loading dose may be needed at the onset of therapy.
- This is calculated as follows:

$$\text{Loading dose} = \frac{V_d \times TC}{F}$$

- When the drug is given by constant IV infusion the $TC = C_{ss}$

CASE 1. CONTINUOUS IV INFUSION

- constant IV infusion
- constant IV infusion with loading dose



WRAPPING UP SO FAR

$$\text{Dosing rate} = \frac{CL \times TC}{F}$$

$$\text{Loading dose} = \frac{V_d \times TC}{F}$$

$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

WRAPPING UP SO FAR

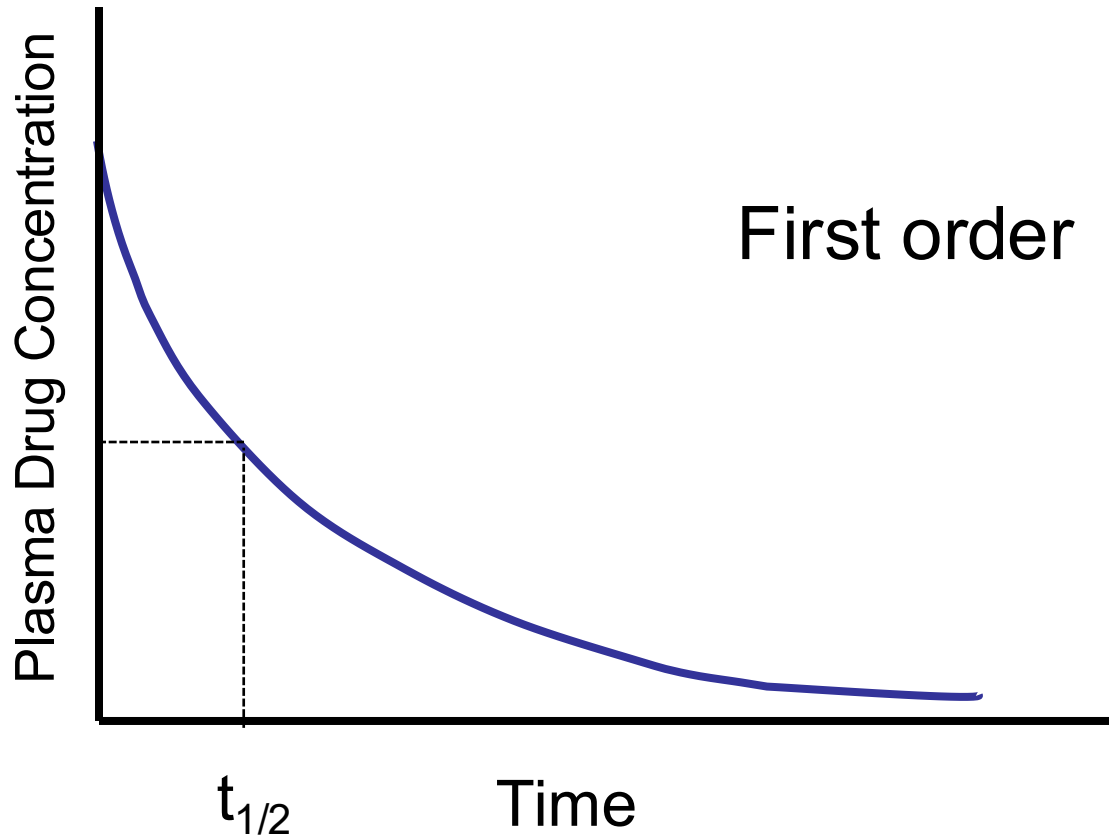
- The clearance determines the maintenance dosing rate.
- The volume of distribution determines the loading dose.
- The half-life determines the time to steady state and the dosing interval.

CASE 2. FIXED-DOSE, FIXED-TIME INTERVAL REGIMENS

- Fixed doses result in fluctuations in the plasma level of the drug.

CASE 2A. Single IV Injection

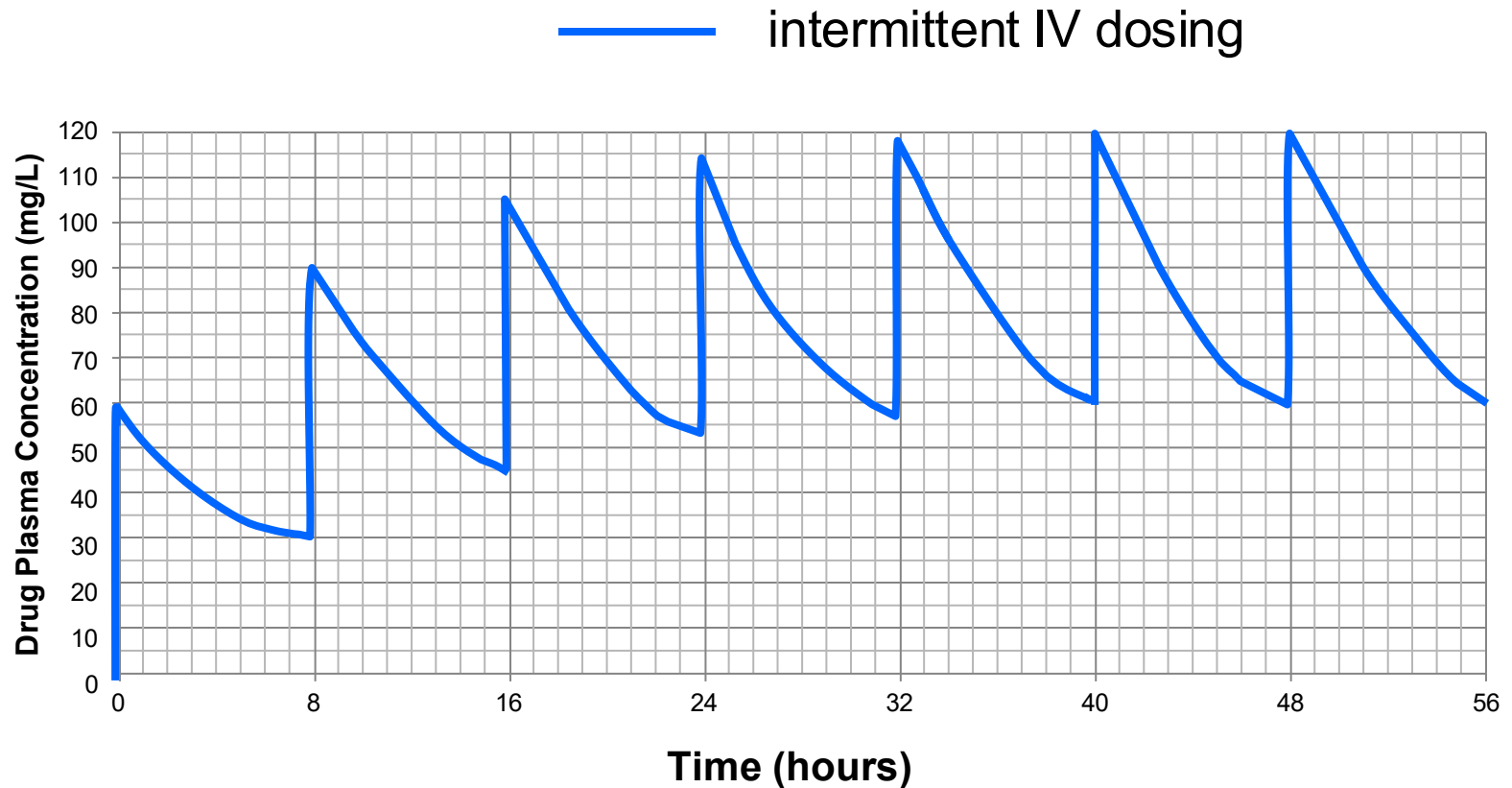
- The plasma level will peak and decrease exponentially with time.



CASE 2B. Multiple IV Injections

- When the drug is given repeatedly at regular intervals, plasma concentration increases until a steady state is reached.
- The plasma concentration fluctuates about a mean.

CASE 2B. Multiple IV Injections



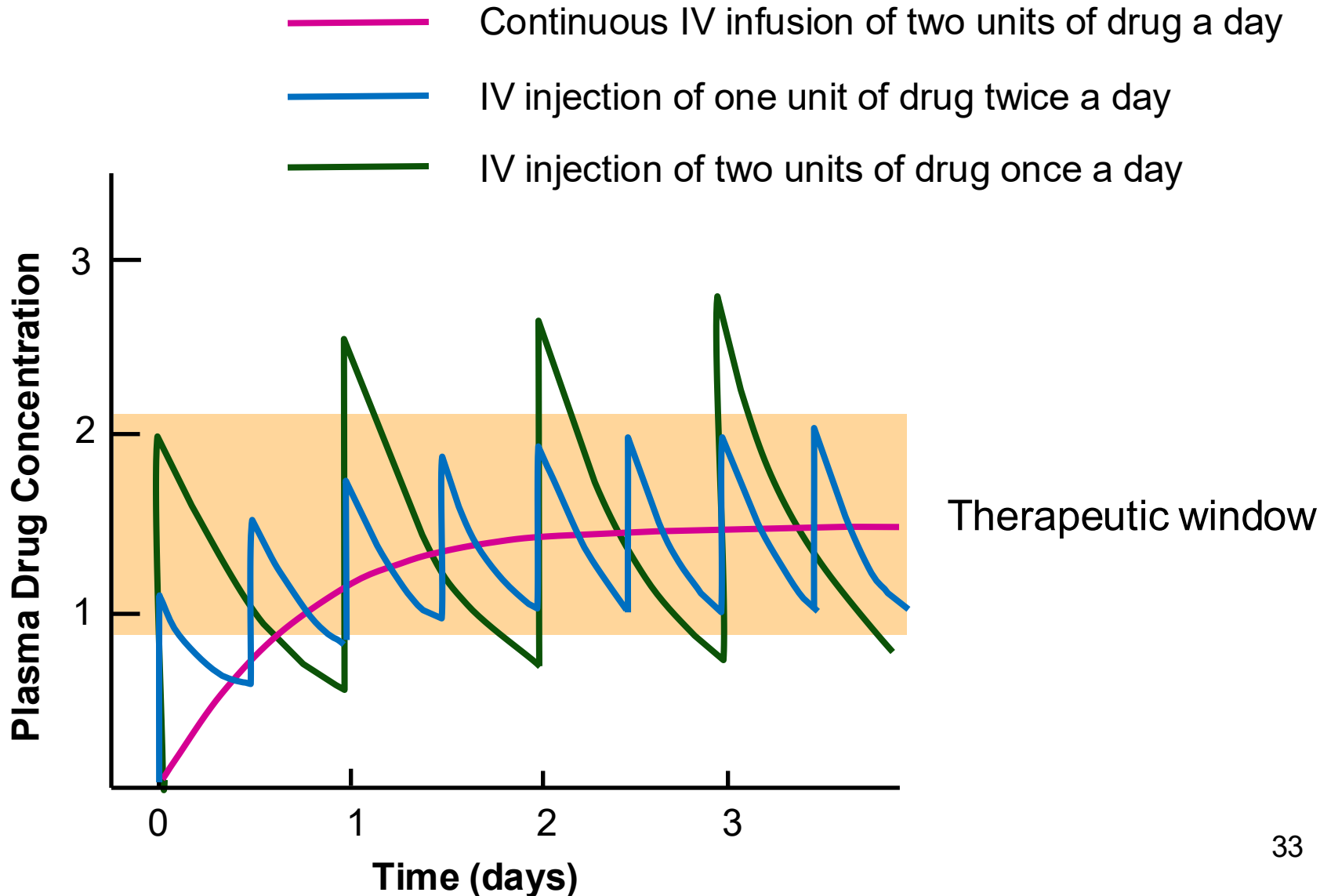
CASE 2B. Multiple IV Injections

- Giving smaller doses at shorter intervals reduces the amplitude of the swings in drug concentration.
- The number of doses to be given per day is determined by the half-life of the drug and the therapeutic window.

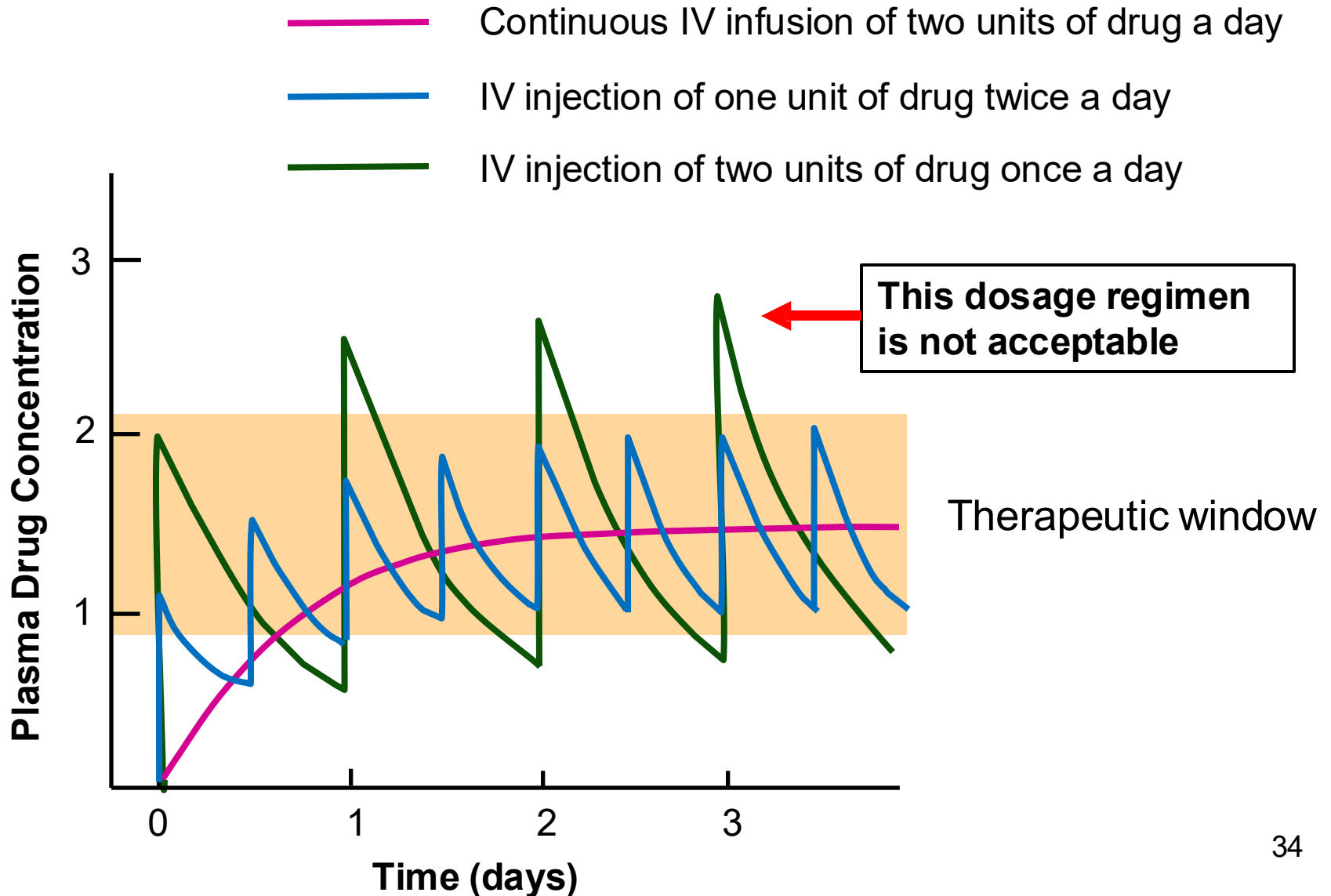
CASE 2B. Multiple IV Injections

- If the therapeutic window is narrow, then smaller, more frequent doses must be administered to avoid toxicity.
- The drug C_{ss} and the rate at which steady state is approached are not affected by dosing frequency.

CASE 2B. Multiple IV Injections



CASE 2B. Multiple IV Injections



DRUG ACCUMULATION

- Whenever drug doses are repeated, the drug will accumulate in the body until dosing stops.
- Accumulation is inversely proportional to the fraction of the dose lost in each dosing interval.

DRUG ACCUMULATION

- A convenient index of accumulation is the accumulation factor (AF).

$$AF = \frac{1}{\text{Fraction lost in one dosing interval}}$$

DRUG ACCUMULATION

- The AF predicts the ratio of the peak concentration at steady state to the peak concentration after the first dose.
- For a drug given once every half-life:

$$AF = 1/0.5 = 2$$

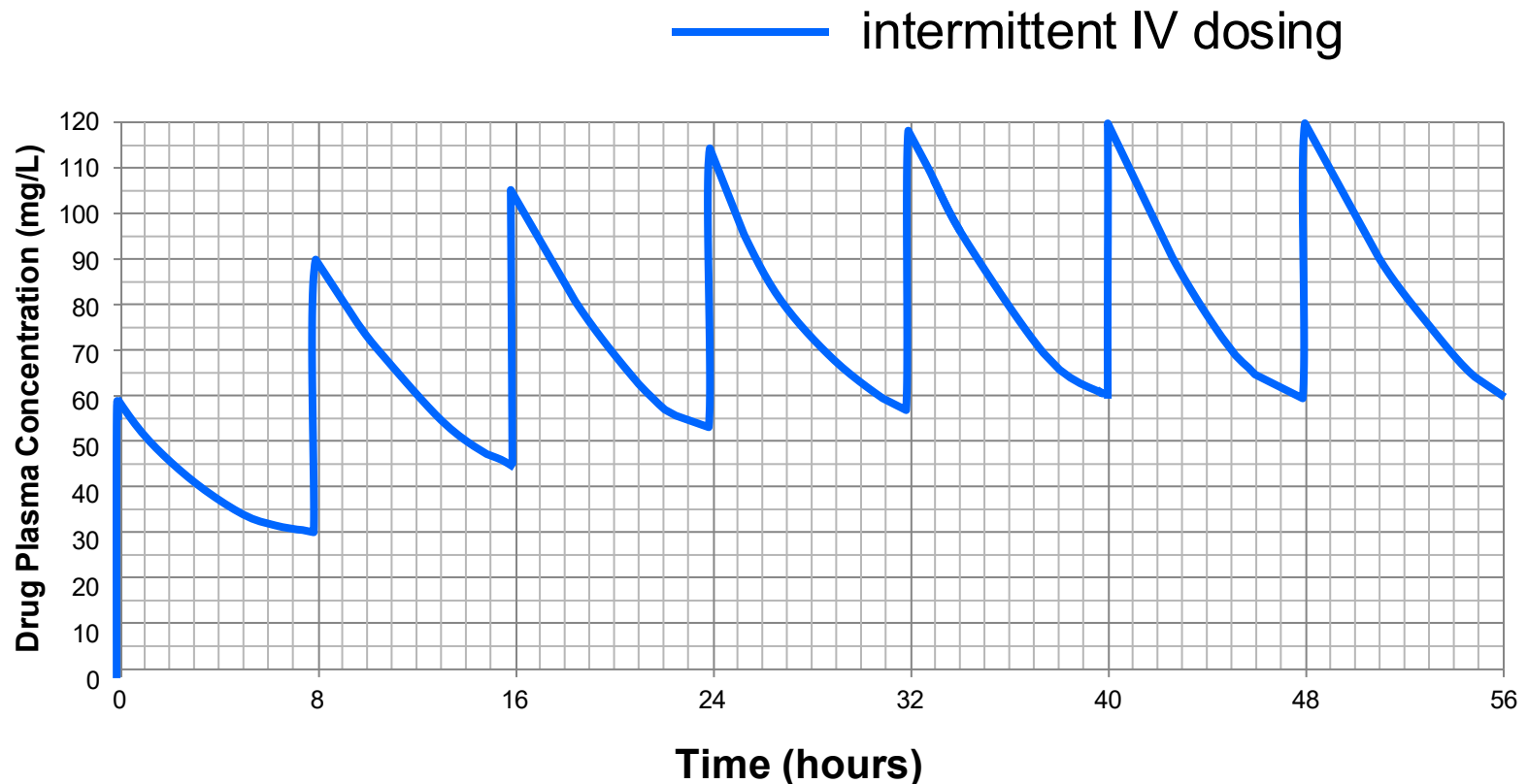
DRUG ACCUMULATION

- The peak concentrations at steady-state will be equal to the peak concentration after the first dose multiplied by the AF.

$$PC_{ss} = PC1 \times AF$$

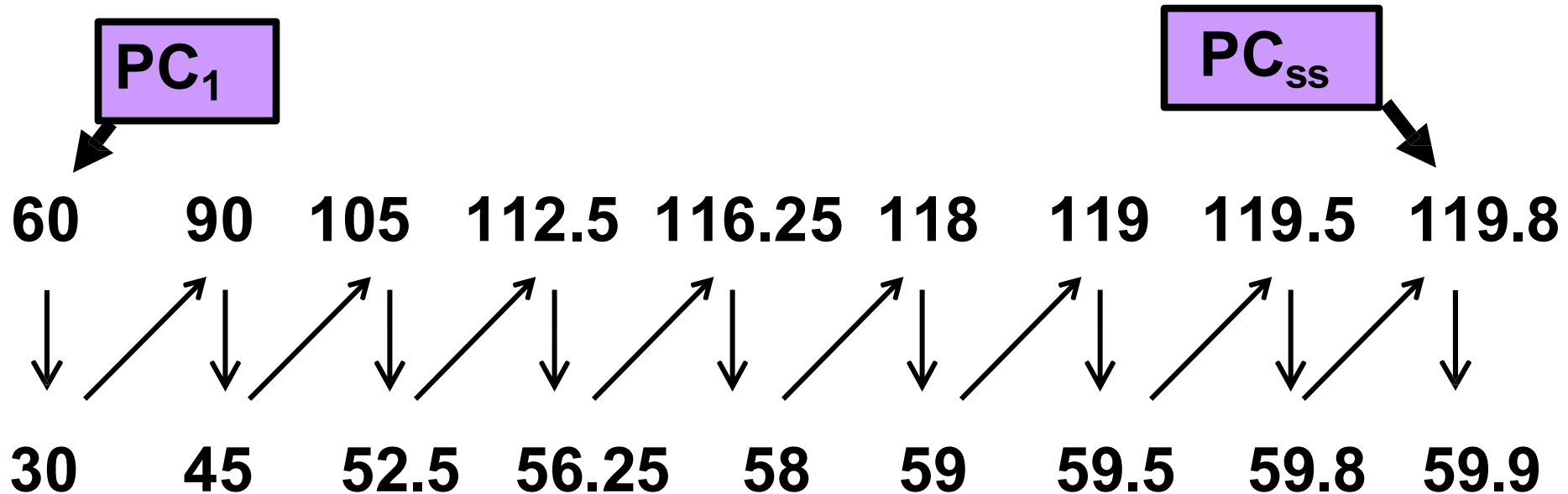
DRUG ACCUMULATION: EXAMPLE

- For a drug given once every half-life the $AF = 2$. Therefore the $PC_{ss} = 2 \times PC_1$.



DRUG ACCUMULATION: EXAMPLE

A drug given once every half-life. $AF = 2$



LOADING DOSE WITH INTERMITTENT DOSING

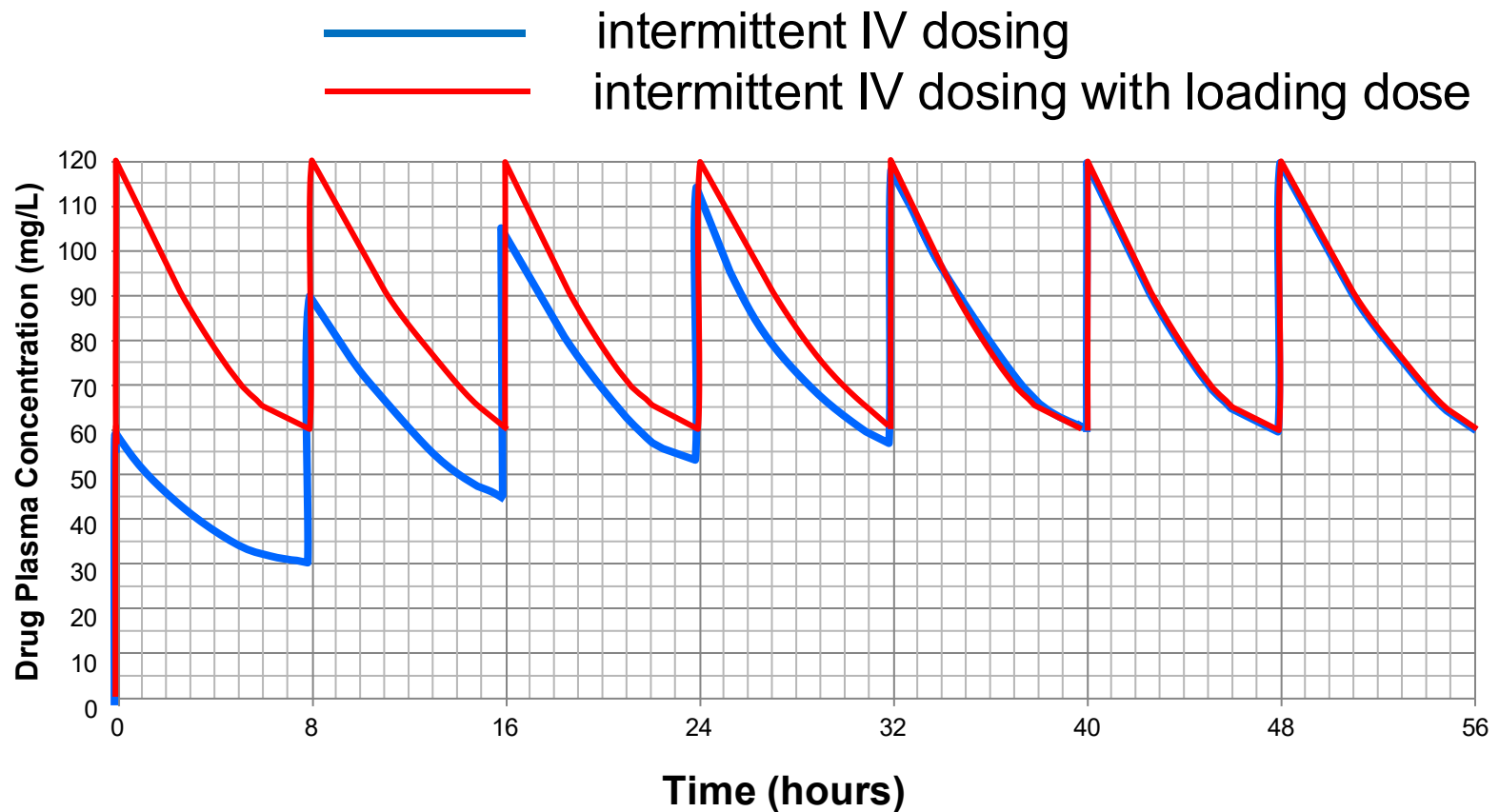
- When intermittent doses are given, the TC in the equation

$$LD = V_d \times TC$$

Should be the peak concentration at steady state (PC_{ss}).

$$LD = V_d \times PC_{ss}$$

CASE 2B. Multiple IV Injections



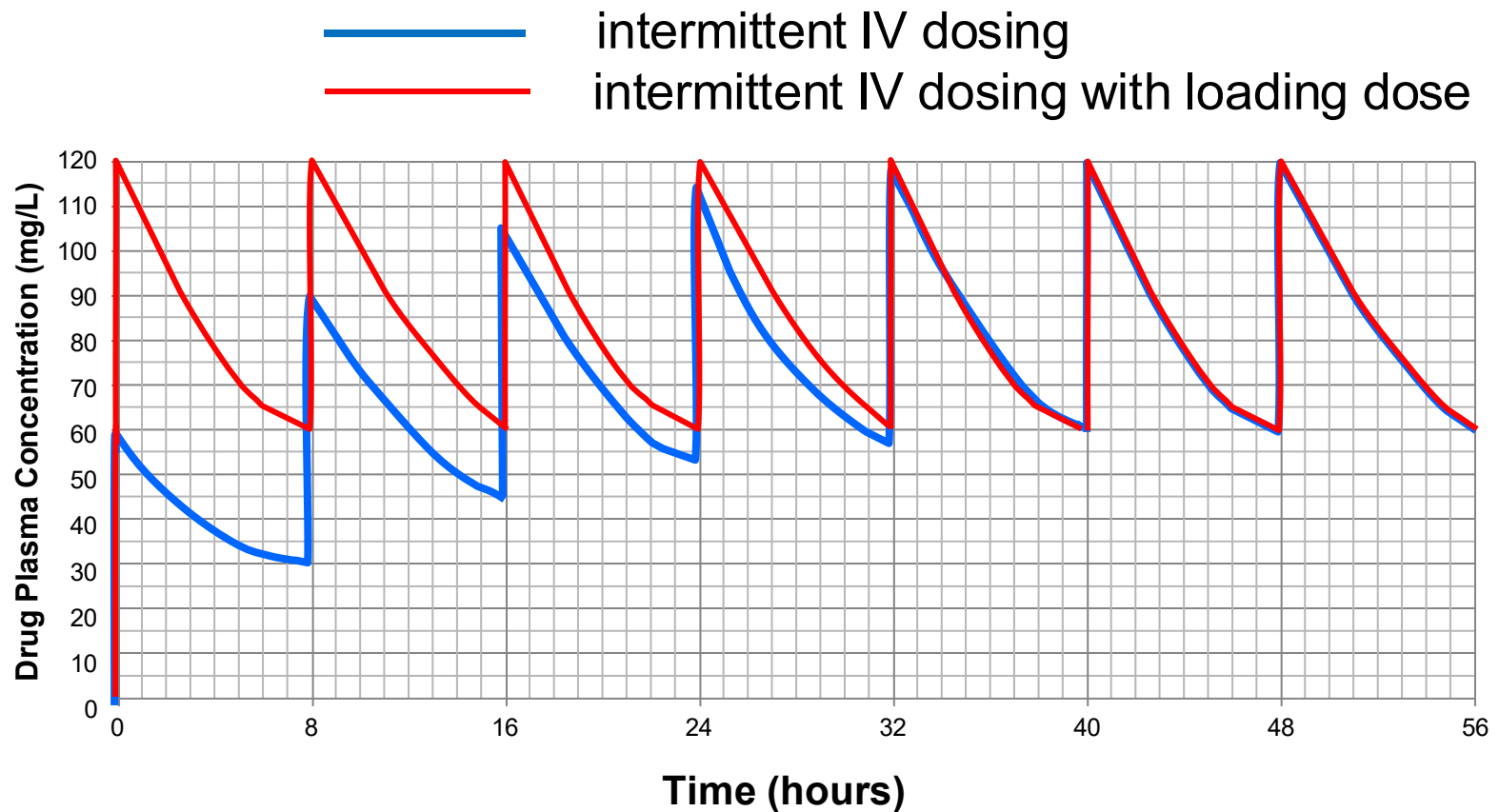
LOADING DOSE WITH INTERMITTENT DOSING

- If we don't know the PCss, we can calculate the LD using the accumulation factor.

LD = Maintenance Dose x Accumulation Factor

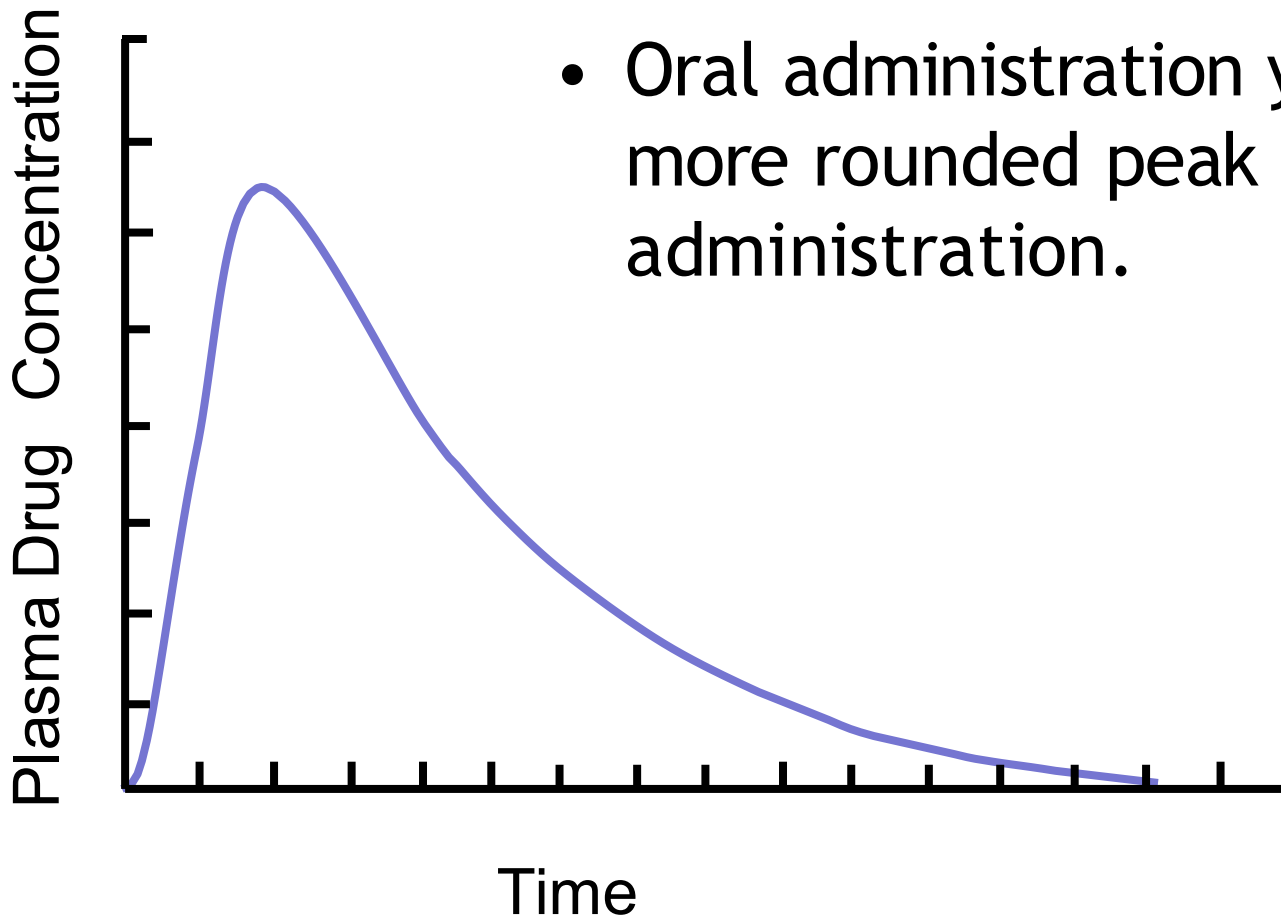
- For a drug given once every half-life the accumulation factor is 2.
- Therefore, the loading dose should be twice the maintenance dose.

CASE 2B. Multiple IV Injections



CASE 2C. Orally Administered Drugs

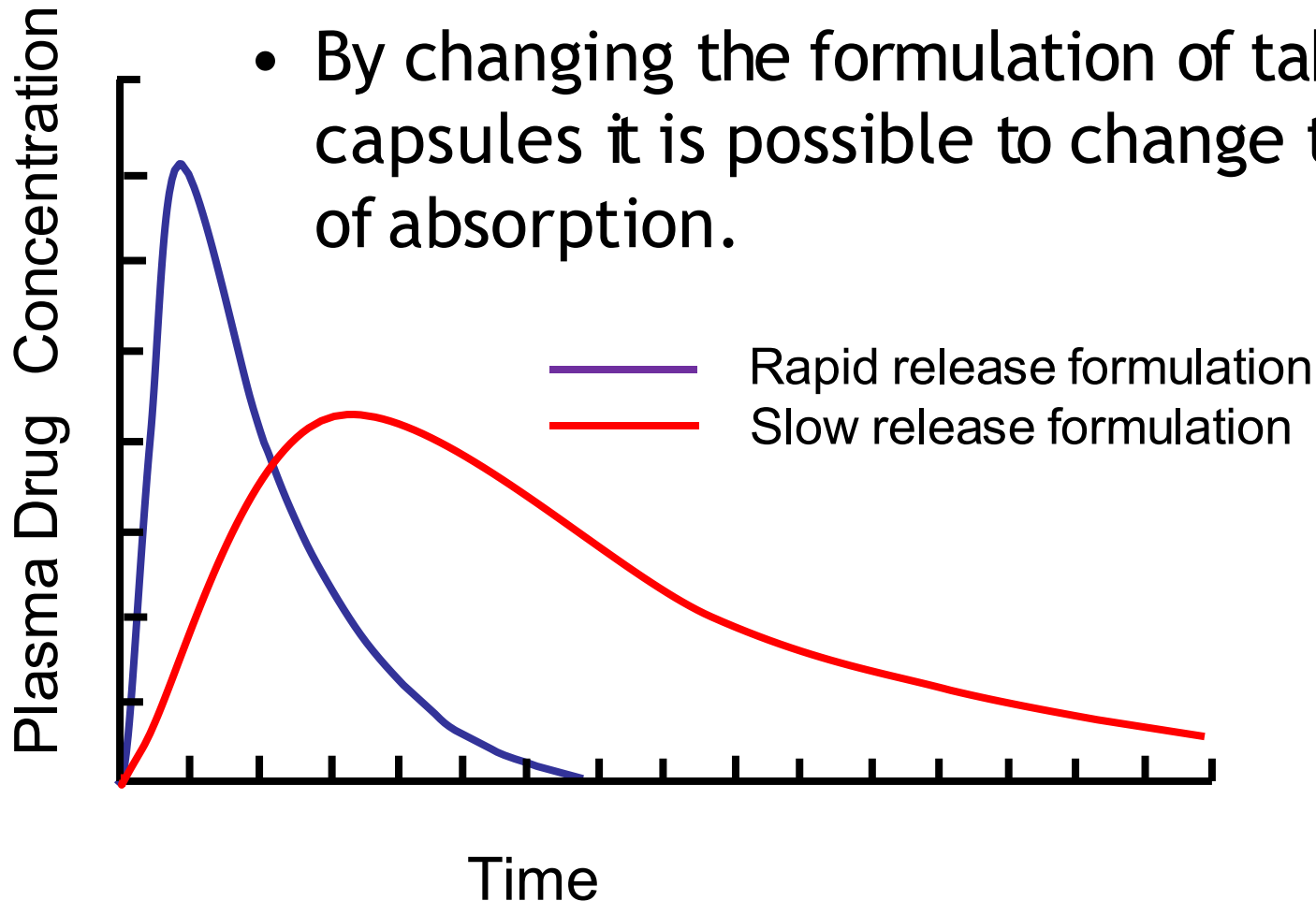
Single oral administration



CASE 2C. Orally Administered Drugs

Changing rate of absorption

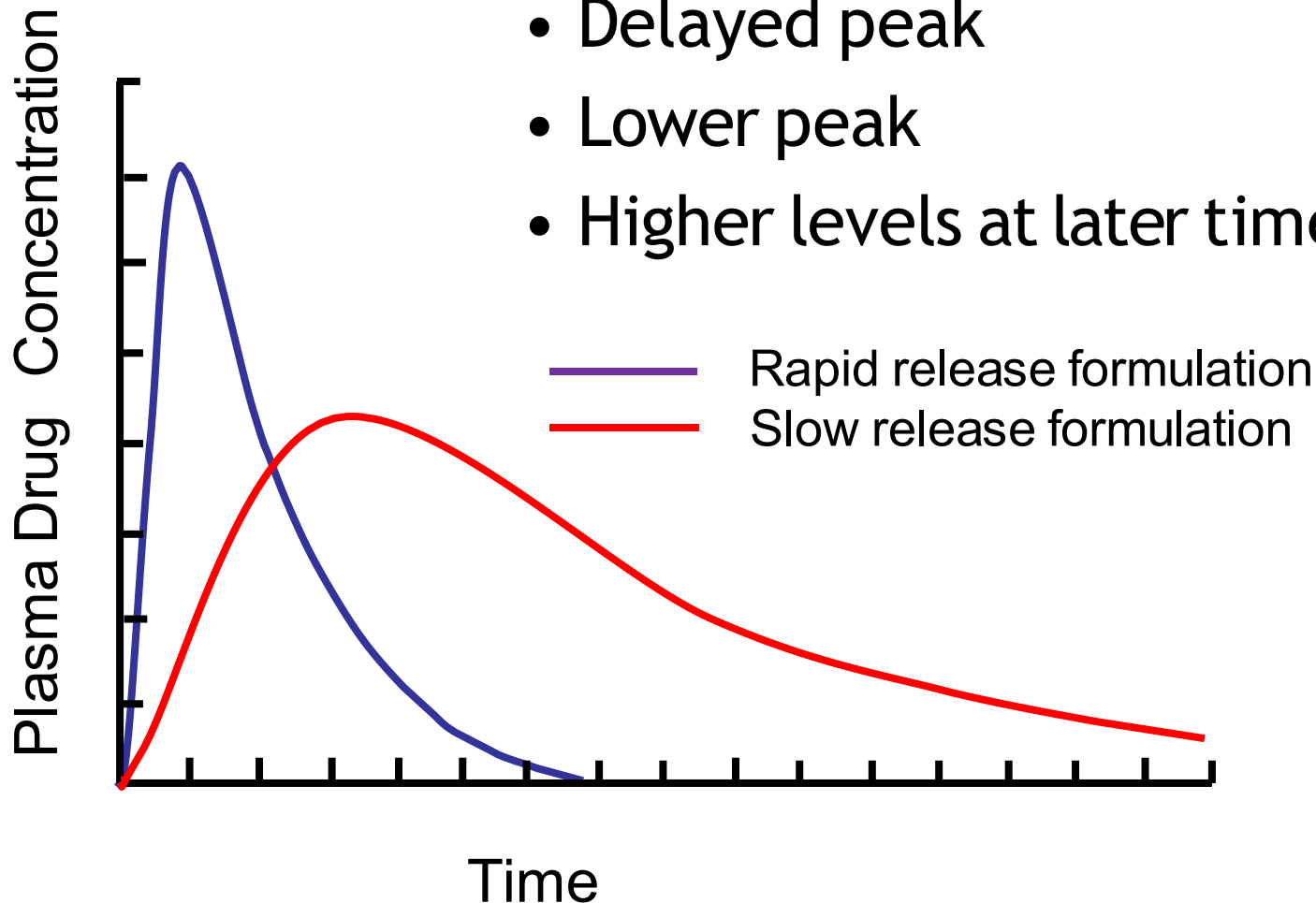
- By changing the formulation of tablets or capsules it is possible to change the rate of absorption.



CASE 2C. Orally Administered Drugs

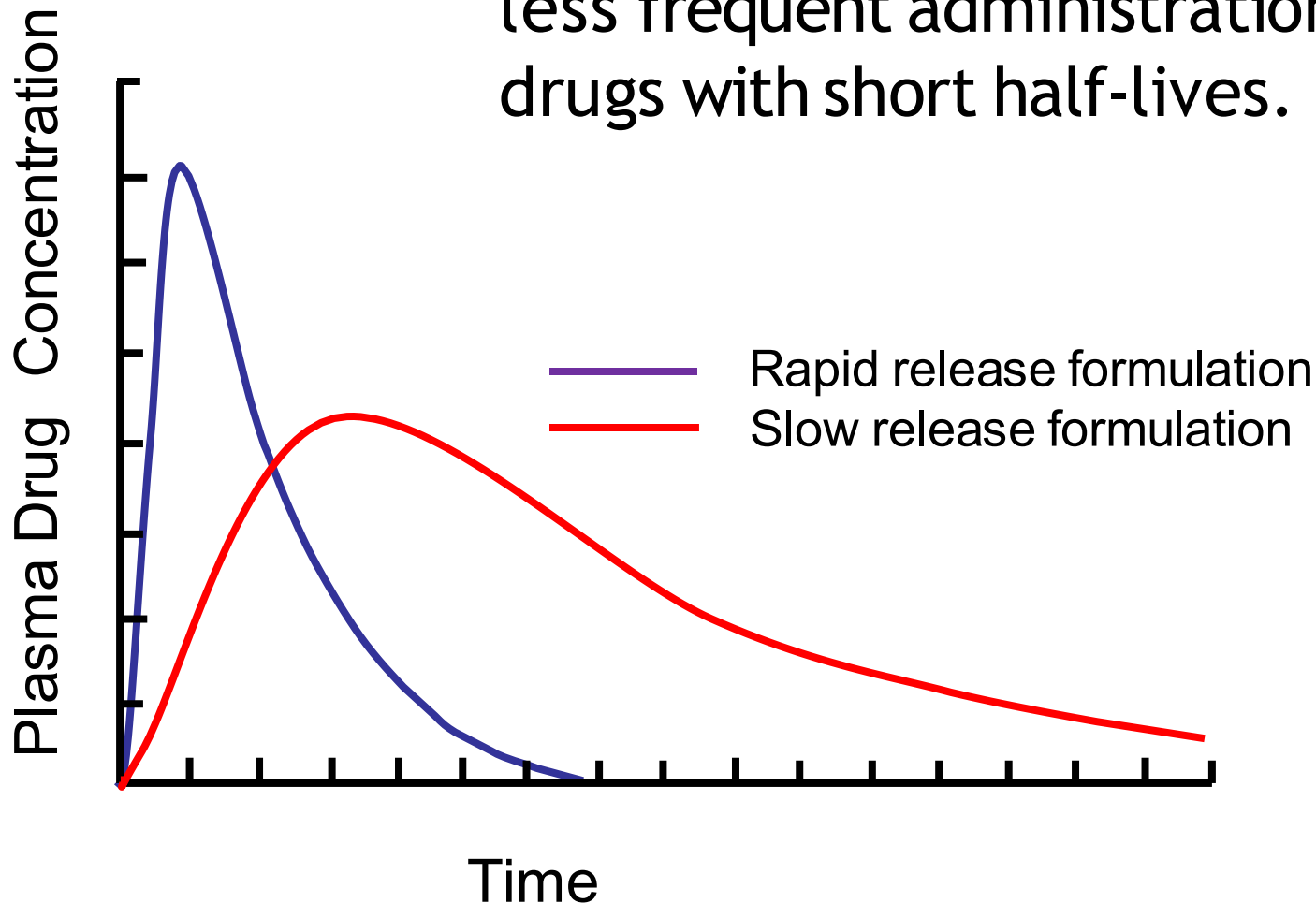
Slow release formulation:

- Delayed peak
- Lower peak
- Higher levels at later times



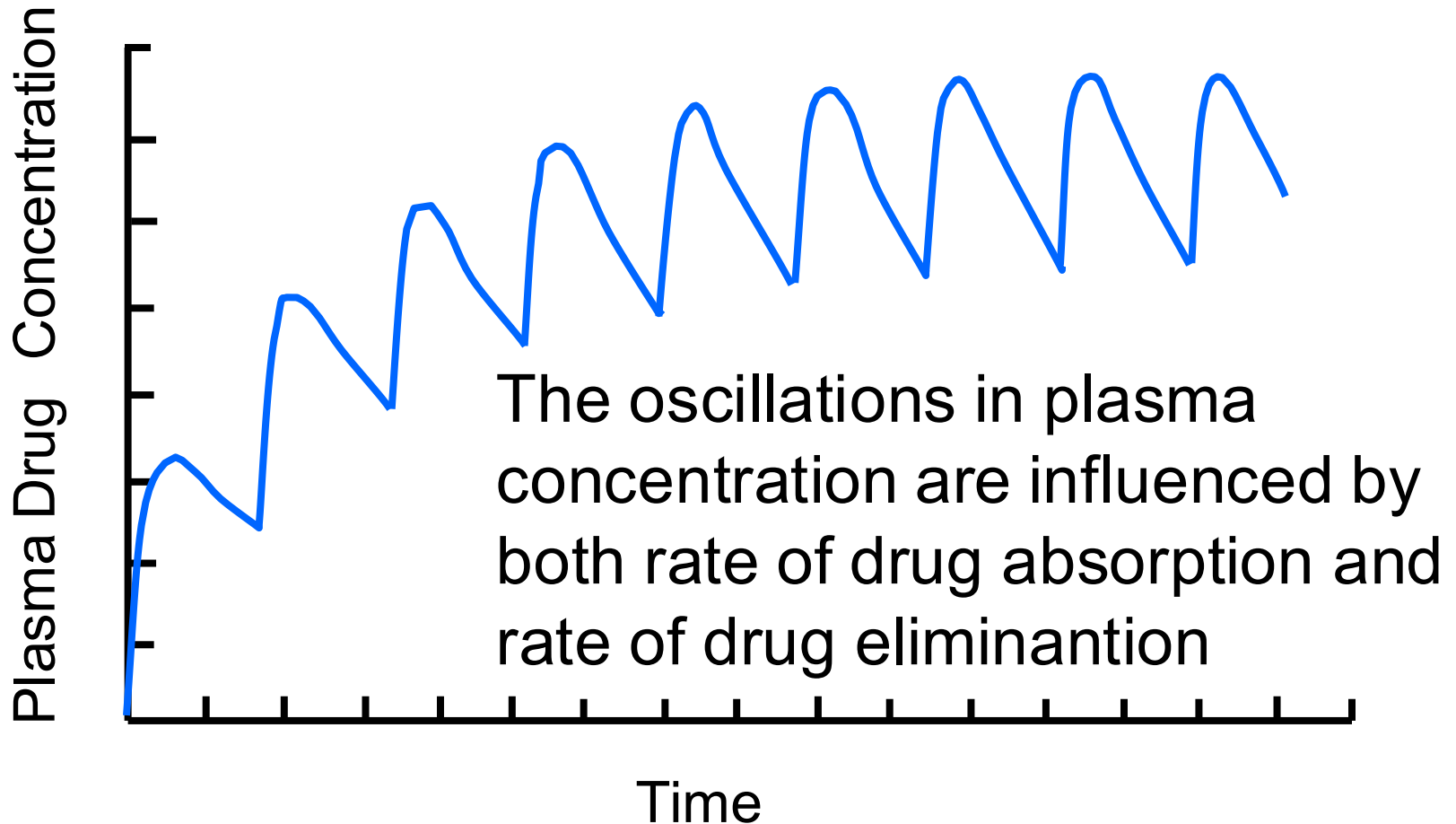
CASE 2C. Orally Administered Drugs

- Slow release formulations allow less frequent administration of drugs with short half-lives.

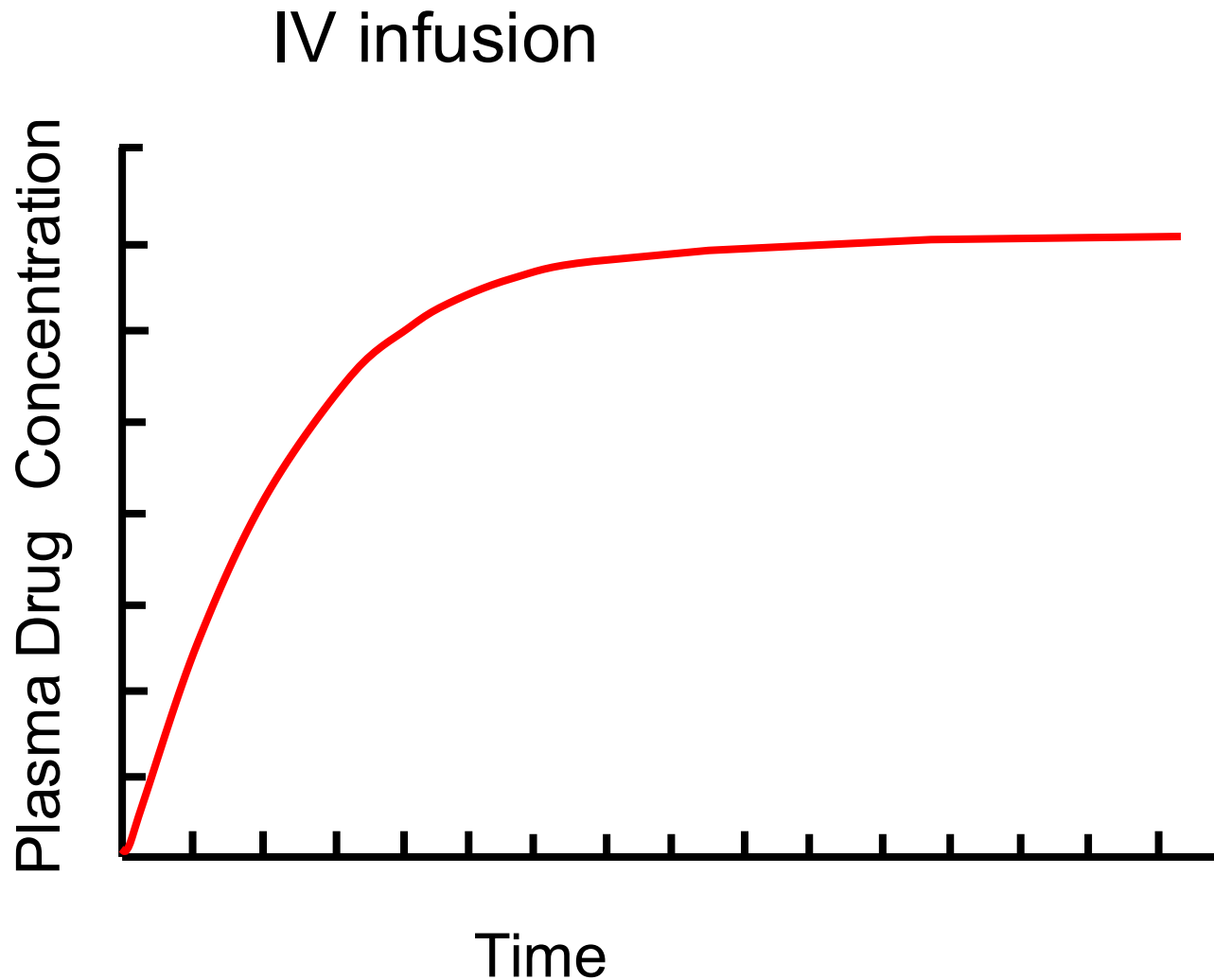


CASE 2C. Orally Administered Drugs

Multiple oral administration

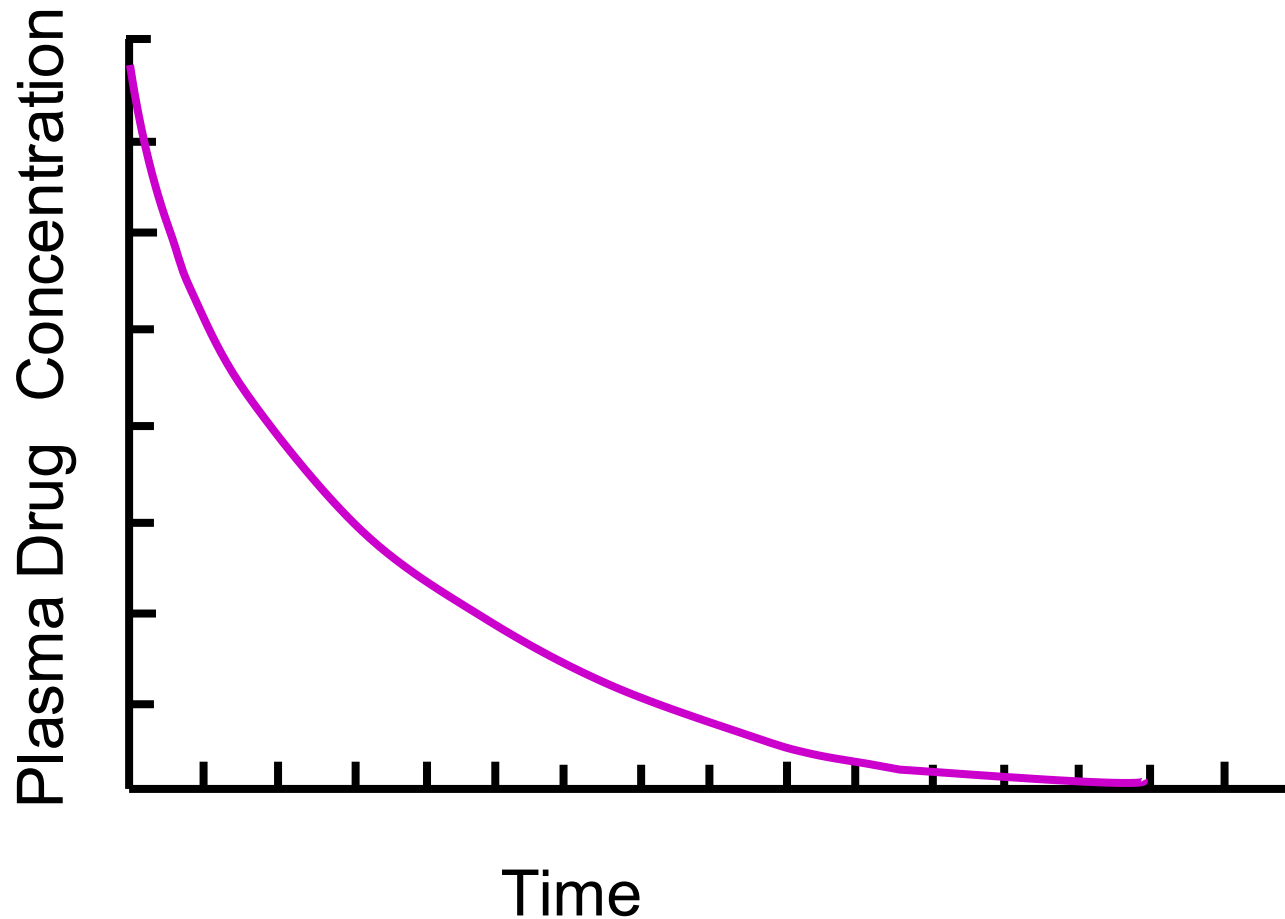


TIME-COURSE OF PLASMA CONCENTRATION



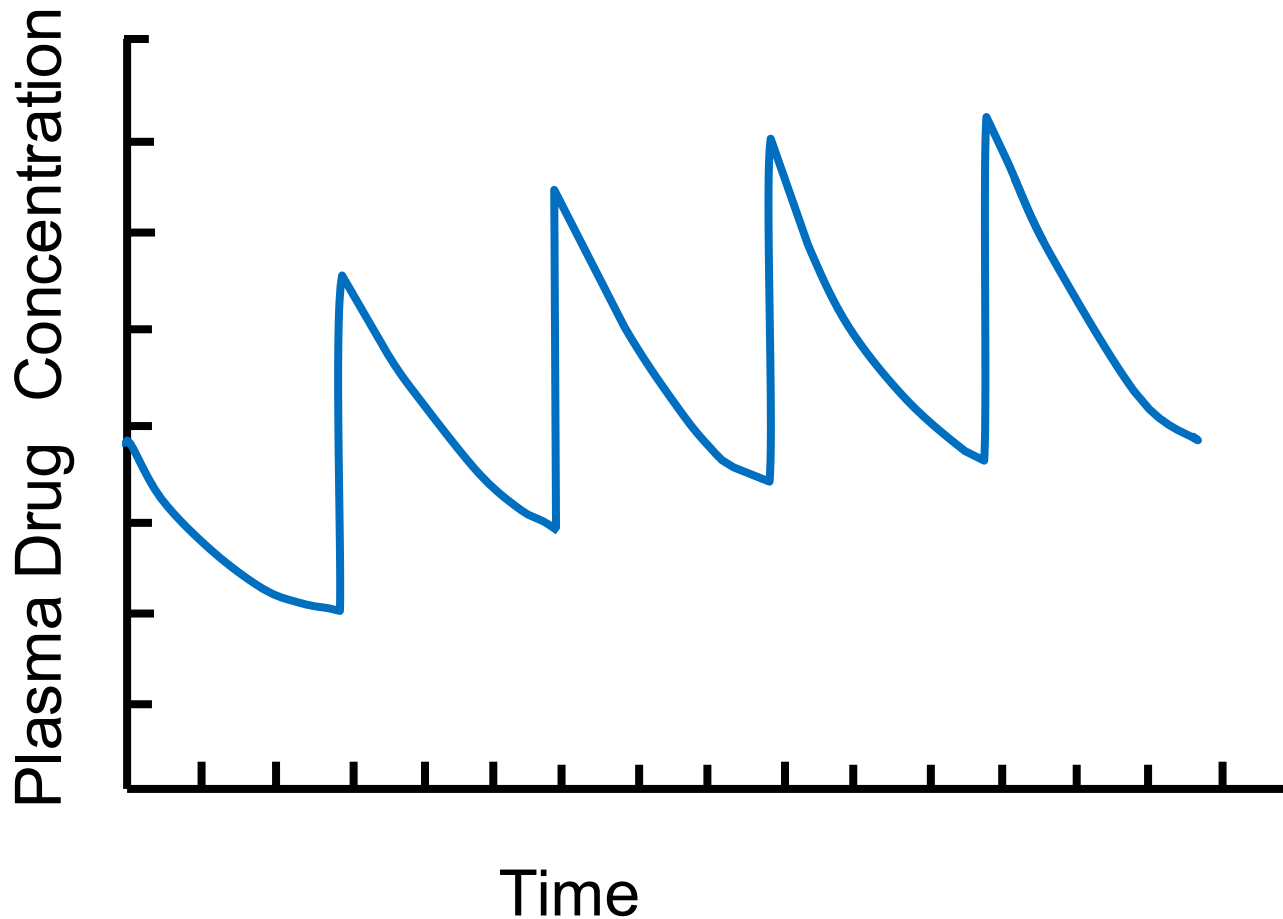
TIME-COURSE OF PLASMA CONCENTRATION

IV bolus injection



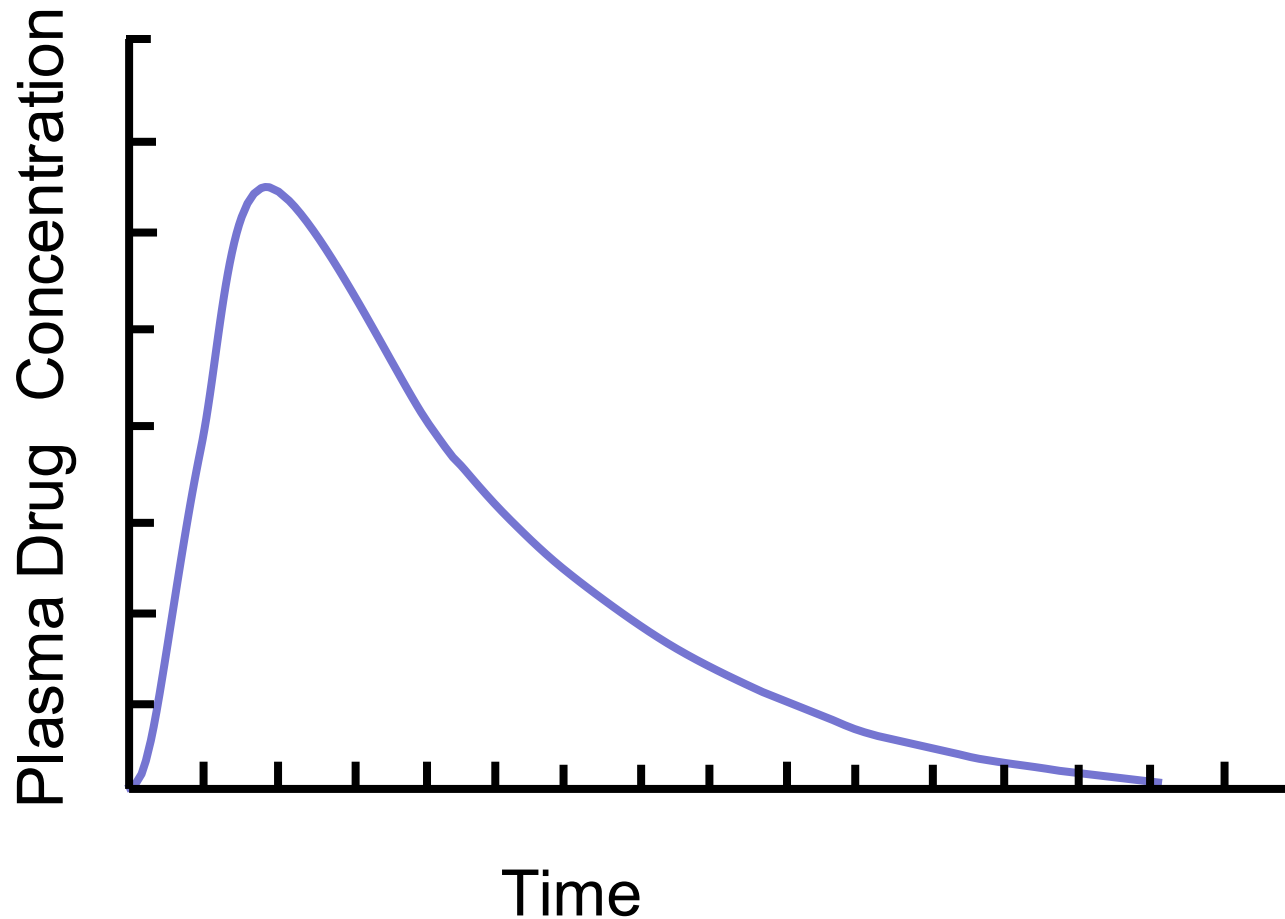
TIME-COURSE OF PLASMA CONCENTRATION

Multiple IV injections



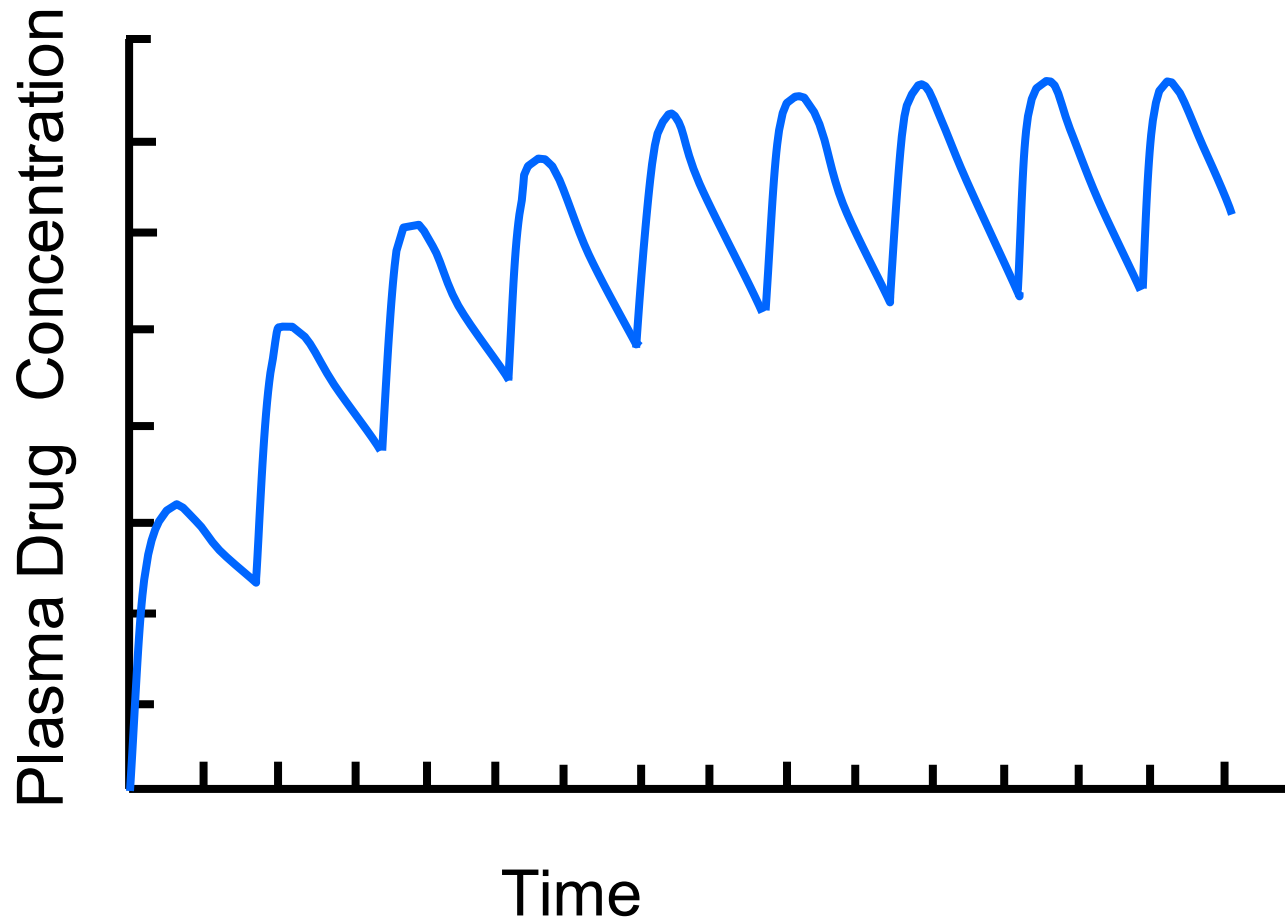
TIME-COURSE OF PLASMA CONCENTRATION

Single oral administration



TIME-COURSE OF PLASMA CONCENTRATION

Multiple oral administrations



Example: Maintenance Dose Calculation

- A target plasma theophylline concentration of 10 mg/L is desired to relieve acute bronchial asthma in a patient.
- From the tables we obtain the mean clearance value for theophylline: 2.8 L/h/70 Kg.
- What should the **dosing rate** be?

Example: Maintenance Dose Calculation

- The drug will be given as IV infusion, so $F = 1$
- **Dosing rate = $CL \times C$**
- Dosing rate = $2.8 \text{ L/h/70 Kg} \times 10 \text{ mg/L} = 28 \text{ mg/h/70 Kg}$
- Therefore, in this patient, the infusion rate should be 28 mg/h/70 Kg

Example: Maintenance Dose Calculation

- If the asthma attack is relieved, the clinician might want to maintain this plasma level using oral theophylline.

Example: Maintenance Dose Calculation

- Oral theophylline might be given **every 12 hours** using an extended release formulation to approximate a continuous IV infusion.
- According to tables $F_{\text{oral}} = 0.6$.

Example: Maintenance Dose Calculation

- When the dosing interval is 12 hours, the size of each maintenance dose would be:

$$\text{Dose} = \frac{\text{dosing rate}}{F} \times \text{dosing interval}$$

$$= \frac{28 \text{ mg/h}}{0.96} \times 12 \text{ hours} = 350 \text{ mg}$$

- A 350-mg tablet would be prescribed at 12-h intervals.

HIGH-YIELD PHARMACOKINETICS EQUATIONS

$\text{pH} - \text{pK} = \log \frac{[\text{Unprotonated form}]}{[\text{Protonated form}]}$	$\text{CL} = \frac{\text{Rate of elimination of drug}}{\text{Plasma drug concentration}}$
$\text{Dosing rate} = \frac{\text{CL} \times \text{TC}}{\text{F}}$	$\text{V}_d = \frac{\text{Amount of drug in the body}}{\text{Plasma drug concentration}}$
$t_{1/2} = \frac{0.693 \times \text{V}_d}{\text{CL}}$	$\text{Loading dose} = \frac{\text{V}_d \times \text{TC}}{\text{F}}$
$\text{Maintenance dose} = \frac{\text{Dosing rate}}{\text{F}} \times \text{Dosing interval}$	
$\text{AF} = \frac{1}{\text{Fraction lost in one dosing interval}}$	$\text{PC}_{ss} = \text{PC}_1 \times \text{AF} \text{ (intermittent dosing)}$ $\text{LD} = \text{MD} \times \text{AF} \text{ (intermittent dosing)}$



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